

## Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

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|---------------|---------------------------------------------------------------------------------------|
| TEAM ID       | SWTID-2026-6767                                                                       |
| PROJECT TITLE | Visualization Tool for Electric Vehicle<br>Charge and Range Analysis using<br>Tableau |
| MAX MARKS     | 5 MARKS                                                                               |

### CHAPTER – IV PROJECT DESIGN

#### 4.1 – Problem Solution Fit

| Problem                                             | Solution                                                                                             |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Difficulty understanding battery performance data   | Interactive Tableau dashboards provide clear visual insights into battery status and health.         |
| Range anxiety during long-distance travel           | Range analysis and forecasting dashboards help users estimate remaining driving distance accurately. |
| Lack of visibility into charging patterns           | Charging history visualizations show charging frequency, duration, and efficiency.                   |
| Difficulty locating suitable charging opportunities | Map-based visualizations display charging station locations and accessibility.                       |
| Inefficient energy consumption analysis             | Energy usage dashboards identify driving behaviours that affect battery performance.                 |
| Complex raw EV datasets                             | Tableau converts complex data into easy-to-understand charts, graphs, and reports.                   |
| Limited support for data-driven trip planning       | Predictive analytics and range forecasting support better travel decisions.                          |